MRPL REFINERY STANDARD OPERATING PROCEDURE
SOP-402: Emergency Isolation of Crude Distillation Feed Pump P-204A
Plant Unit: CDU-1 | Revision: 3.2

1.0 Objective
This procedure specifies mandatory safety isolation steps for Centrifugal Pump P-204A prior to mechanical seal

2.0 Hazard Warnings
WARNING: Heavy crude oil is maintained at 180 deg C and 14.5 bar operating pressure.
Failure to isolate upstream block valve may result in high-pressure hydrocarbon release.

3.0 Isolation Sequence Steps
Step 1: Notify the Central Control Room (CCR) board operator before initiating field isolation.
Step 2: Press local emergency stop push button (PB-204) and verify electric motor de-energization.
Step 3: Close suction block valve V-101 (8-inch gate valve) and apply Lockout/Tagout (LOTO) lock # CDU1-LOT
Step 4: Close discharge block valve V-102 (6-inch globe valve) and secure with mechanical chain.
Step 5: Open casing drain valve DV-01 to flare header to depressurize the pump casing to 0 barg.

4.0 Post-Isolation Verification
Step 6: Verify pressure gauge PG-204 reads exactly 0.0 kg/cm2.
Step 7: Perform combustible gas test at pump seal housing with calibrated multigas detector. Hydrocarbon conc
Step 8: Sign section B of the Cold Work Permit and hand over to Mechanical Maintenance Supervisor.

5.0 Normal Operating Parameters for Pump P-204A
- Design Flow Rate: 340 m3/hr
- Discharge Head: 125 meters
- Motor Power: 160 kW, 3.3 kV, 50 Hz
- Lubricant: ISO VG 68 synthetic turbine oil (check oil level every 24 hours).